Defense Trilemma: Geometric Impossibility of Prompt Injection Defense

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Abstract

We prove that no continuous, utility-preserving alignment defense can make a large language model fully safe, and we characterize exactly where every such defense has a gap. Working on a connected Hausdorff space, we establish three impossibility results of increasing strength. *Boundary fixation*: every conforming defense must fix the boundary of the alignment deviation surface. *ε-robust constraint*: when both the defense and alignment function are Lipschitz, a positive-measure band around that boundary remains close to the safety threshold regardless of defense design. *Persistent unsafe region*: under a transversality condition, a positive-measure subset with $f(D(x)) > \tau$ survives every conforming defense. We prove parallel discrete results that require no topology, then extend the theory to multi-turn interactions, stochastic defenses, and capacity-parity settings, showing that the impossibility recurs at every conversational turn, that randomization cannot circumvent the bounds, and that utility preservation taxes the defense even under equal resources. Together these results delineate a *manifold of failure* [18]: a region that persists under every conforming wrapper defense. The full theory is mechanically verified in Lean 4 with Mathlib (40 files, ∼300 theorems, no admitted proofs, three standard axioms) and validated empirically on three LLMs.

1 Introduction

Can you build a wrapper around a language model that eliminates all prompt injection vulnerabilities? Most current work on LLM safety implicitly assumes the answer is yes, at least in principle. The defense mechanisms proposed span a wide range: input classifiers that flag suspicious prompts [1], output filters that catch unsafe completions [8], and constitutional rewriting pipelines that reshape model behavior [3]. Despite their differences, all share the same underlying structure: they are functions $D\colon X \to X$ that preprocess prompts before the model sees them, aiming to map unsafe inputs to safe equivalents while leaving safe inputs alone.

Prior work has largely focused on discovering or mitigating individual adversarial inputs. We ask a different question: what are the *geometric limits of the defense mapping itself*? We show that under mild and commonly assumed design constraints like continuity and utility preservation, no defense can eliminate the safety boundary. Vulnerabilities arise not from specific inputs, but from the structure of the space and the constraints imposed on admissible defenses. This shifts the problem from adversarial search to geometric constraint.

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We prove the following three results, each strictly stronger than the last:

Boundary fixation (Theorem 4.1). If the defense is continuous, utility-preserving, and the prompt space is connected, then the defense must fix at least one boundary point—a prompt where alignment deviation equals the threshold exactly—passing it through with no remediation.

ε-robust constraint (Theorem 5.1). If the defense is also K-Lipschitz and the alignment deviation function is L-Lipschitz, the defense cannot uniformly reduce alignment deviation far below τ on a neighborhood of the fixed boundary point. For any x within distance δ of the fixed point z:

$$f(D(x)) \;\geq\; \tau - L(K+1)\,\delta. \tag{1}$$

The ε-band $\{x : \tau - \varepsilon \leq f(x) \leq \tau\}$ has positive measure for every $\varepsilon > 0$ (Theorem 5.2). This constrains the *depth* to which the defense can push near-boundary points into the safe region, though it does not by itself prevent the defense from nudging them slightly below τ.

Persistent unsafe region (Theorem 6.3). Under a transversality condition, the alignment surface has directional slope exceeding $L(K+1)$ at the boundary—the defense leaves a *positive-measure region that remains unsafe*:

$$f(D(x)) > \tau \quad \text{for all } x \in \mathcal{S}, \qquad \mu(\mathcal{S}) > 0. \tag{2}$$

This is the strongest of the three results: not boundary fixation, not a near-threshold band, but persistent unsafe volume. The condition holds whenever the alignment surface rises faster than the defense can pull it down—empirically observed in all three models we study.

From discrete to continuous. All three results apply to continuous interpolants of discrete data. The Tietze extension theorem guarantees that any finite set of behavioral observations in a normal space admits continuous extensions, and the impossibilities hold for every such extension (Theorem 7.1). This does not claim the results hold for the underlying discrete token system directly, but for every continuous model consistent with the observations.

Both the basic and ε-robust proofs are machine-verified in Lean 4 with zero `sorry` statements, as is the persistent unsafe region theorem. All three conditions—continuity, utility preservation, and connectedness—are individually necessary; we give counterexamples for each (Section A).

2 Related Work

Adversarial robustness. The existence of adversarial examples in neural networks has been extensively studied [21, 7, 11, 15], along with a large body of work on certified robustness via abstract interpretation [20], SMT-based verification [9, 14], and randomized smoothing [6]. These approaches characterize the existence of inputs that induce failure, or provide guarantees within bounded perturbation sets. Our work differs from this paradigm: rather than analyzing which inputs cause failures for a fixed model, we prove that for any continuous, utility-preserving defense D, there exists an input at which the defense fails to enforce safety. This shifts the focus from existential statements over inputs to universal constraints over defenses, independent of perturbation models, norms, or training procedures.

Prompt injection and jailbreaking. Recent work has developed increasingly effective methods for eliciting unsafe behavior from language models, including gradient-based attacks [24], search-based strategies [5, 10], indirect prompt injection [13], and quality-diversity approaches such as MAP-Elites [17, 19]. These methods demonstrate that vulnerabilities are widespread and often easy to discover. In contrast, our results are not algorithmic but structural: existing work shows that failures can be found; we prove that under mild and commonly assumed constraints, failures cannot be eliminated by any defense of the given form. This provides a formal explanation for the empirical persistence of prompt injection: vulnerabilities arise not from insufficient search or imperfect training, but from geometric constraints on admissible transformations of the input space.

Impossibility results in machine learning. Classical results such as the No Free Lunch Theorem [23] show that no algorithm performs optimally across all possible data distributions, while robustness–accuracy tradeoffs [22] demonstrate that improving robustness may degrade standard performance. Fawzi et al. [12] prove that any classifier with sufficient accuracy is inherently vulnerable to adversarial perturbations. Our results are of a different character: we establish pointwise, distribution-free impossibility theorems that hold for any fixed task and any admissible defense mapping. The limitation arises not from competing objectives or unknown distributions, but from the interaction between continuity, utility preservation, and the topology of the input space. As such, our results are orthogonal to, and not implied by, prior tradeoff or no-free-lunch analyses.

Geometric and topological perspectives. Prior work has examined the geometry of decision boundaries and the topology of learned representations [16], highlighting the importance of structure in understanding model behavior. While our proofs rely on elementary topological arguments, the key novelty lies in identifying that standard design constraints for LLM defenses induce a forced fixed-point structure at the safety boundary. To our knowledge, this connection—linking utility preservation and continuity to unavoidable boundary fixation—has not been previously formalized. Our result further strengthens this perspective by showing that under metric assumptions, unsafe behavior persists over regions of positive measure, not merely at isolated boundary points.

Formal verification of machine learning systems. Formal methods have been used to verify properties of neural networks and learning algorithms [9, 20, 14, 2], typically providing guarantees for specific architectures under bounded input domains. In contrast, our work establishes impossibility results at the level of abstraction of defense functions, independent of any particular model, architecture, or verification procedure. The use of Lean 4 in our work serves to mechanize these general results, rather than to certify properties of a specific system.

3 Formal Setup

Before defining the model formally, we set up the intuition. Imagine that every possible prompt lives in a space X. A scoring function f assigns each prompt a number measuring how far the model's response deviates from its alignment goals: low scores are safe, high scores are unsafe. A fixed threshold τ draws the line between the two. A *defense* is a preprocessor that rewrites prompts before the model sees them, trying to push unsafe prompts into safe territory. We require two natural properties: *continuity* (similar prompts produce similar rewrites, i.e., no brittle edge-case behavior) and *utility preservation* (safe prompts pass through unchanged, so the defense does not degrade normal use).

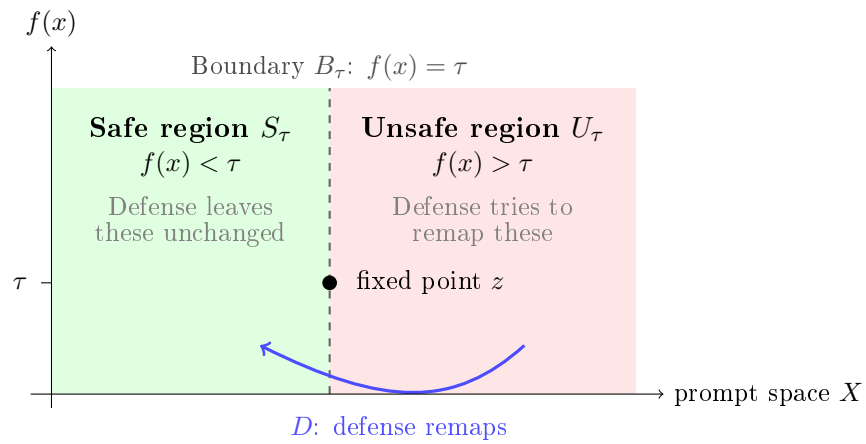

Figure 1: Schematic of the prompt space. The defense D must leave all safe inputs unchanged (utility preservation) and tries to remap unsafe inputs into the safe region. The boundary fixation theorem proves that the defense must also leave some boundary points unchanged—the black dot z passes through with no remediation.

Figure 1 illustrates the setup: the prompt space is divided into a safe region (green), an unsafe region (red), and a boundary (dashed) where the score equals the threshold exactly.

Definition 3.1 (Alignment Deviation Function)**.** *Let X be a topological space (a set equipped with a notion of "nearness" between points). An* alignment deviation function *is a continuous map $f\colon X \to \mathbb{R}$—that is, small changes in the input produce small changes in the score. Given a threshold $\tau \in \mathbb{R}$:*

$$S_\tau = \{x \in X : f(x) < \tau\} \qquad \textit{(safe region)} \tag{3}$$

$$U_\tau = \{x \in X : f(x) > \tau\} \qquad \textit{(unsafe region)} \tag{4}$$

$$B_\tau = \{x \in X : f(x) = \tau\} \qquad \textit{(boundary)} \tag{5}$$

Definition 3.2 (Defense)**.** *A* defense *is a continuous map $D\colon X \to X$. It is* utility-preserving *if $D(x) = x$ for all $x \in S_\tau$ (safe prompts pass through unchanged), and* complete *if $f(D(x)) < \tau$ for all $x \in X$ (every output is safe).*

The central question is: can a defense be both utility-preserving and complete? The following sections prove that, under natural conditions, the answer is no.

4 Boundary Fixation

Intuition. Think of the safe region as a green zone and the unsafe region as a red zone, with a boundary line between them. The defense promises: "I won't touch anything in the green zone." But the green zone doesn't include its own edge—it's defined by a strict inequality ($<$, not $\leq$). So the defense's "don't touch" commitment, which must extend continuously to the edge, includes some points *on* the boundary line. Those boundary prompts—right at the safety threshold—pass through the defense without remediation.

Formal argument. A utility-preserving defense must leave all safe inputs unchanged—it fixes them in place. But continuity forces the set of fixed points to be *closed* (it contains its own limit

points): you cannot "stop fixing" inputs without a discontinuous jump. Meanwhile, the safe region is *open* (it does not contain its boundary): it is defined by a strict inequality $f(x) < \tau$. In a connected space (one that cannot be split into two separate pieces), an open set that is also nonempty and proper cannot be closed. So the defense's fixed-point set, which contains the entire open safe region, must extend past it—into the boundary where $f(z) = \tau$ exactly. The defense must pass these boundary prompts through without remediation.

Theorem 4.1 (Boundary Fixation)**.** *Let X be a connected Hausdorff space (a space that is "in one piece" and where distinct points can be separated by neighborhoods). Let $f\colon X \to \mathbb{R}$ be continuous with $S_\tau \neq \emptyset$ and $U_\tau \neq \emptyset$. Let $D\colon X \to X$ be continuous with $D|_{S_\tau} = \mathrm{id}$.*

Then there exists $z \in X$ with $f(z) = \tau$ and $D(z) = z$. Moreover, every $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ and $D(z) = z$. This set is nonempty.

Proof. **Step 1** (Hausdorff $\Rightarrow$ fixed-point set is closed). The set $\mathrm{Fix}(D) = \{x : D(x) = x\}$ is the preimage of the diagonal $\Delta \subset X \times X$ under the continuous map $x \mapsto (D(x), x)$. In a Hausdorff space the diagonal is closed, so $\mathrm{Fix}(D)$ is closed.

Step 2 (Utility preservation $\Rightarrow$ safe region $\subseteq$ fixed points). Every safe input satisfies $D(x) = x$, so $S_\tau \subseteq \mathrm{Fix}(D)$. Since $\mathrm{Fix}(D)$ is closed, the closure is also contained: $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$.

Step 3 (Connectedness $\Rightarrow$ safe region is not closed). $S_\tau = f^{-1}((-\infty, \tau))$ is open (preimage of an open set under continuous f). If it were also closed, it would be clopen—but in a connected space, the only clopen sets are $\emptyset$ and X. Since both S_τ and U_τ are nonempty, S_τ is neither, so it is not closed.

Step 4 (Continuity of f $\Rightarrow$ boundary point exists). Since S_τ is not closed, $\overline{S_\tau} \supsetneq S_\tau$: there exists $z \in \overline{S_\tau} \setminus S_\tau$. This point satisfies $f(z) \leq \tau$ (limits of values $< \tau$ cannot exceed τ) and $f(z) \geq \tau$ (since $z \notin S_\tau$). Therefore $f(z) = \tau$: the point lies exactly on the safety boundary.

Step 5 (Combining: defense fixes the boundary point). From Step 2, $z \in \overline{S_\tau} \subseteq \mathrm{Fix}(D)$, so $D(z) = z$. The defense passes z through unchanged, and $f(D(z)) = f(z) = \tau \not< \tau$—the point remains at threshold. □

Corollary 4.2. *No continuous utility-preserving defense is complete.*

All three hypotheses—continuity, utility preservation, and connectedness—are individually necessary. A defense can satisfy at most two of the three (the *defense trilemma*, Figure 2). Counterexamples establishing necessity of each hypothesis appear in Section A.

Remark 4.3 (Not all boundary points are fixed)**.** *The theorem captures $\overline{S_\tau} \setminus S_\tau$, not all of B_τ. Boundary points isolated from the safe side may escape. Verified in Lean with $f(x) = x^2$, $\tau = 0$.*

Remark 4.4 (Lipschitz Bounds in Behavioral Space)**.** *While raw discrete token manipulations are chaotic and inherently non-Lipschitz, our bound L applies to the continuous behavioral manifold constructed via the Tietze extension (Theorem 7.1). In these latent spaces (parameterized by axes like indirection and authority), the alignment surface is remarkably smooth. Our empirical validation on Llama-3-8B (Section 11) yields $L \approx 5$ near the boundary, confirming these Lipschitz constraints have practical predictive power.*

4.1 Relaxed Utility Preservation

A natural objection is that strict utility preservation ($D(x) = x$ for safe x) is too strong: a practical defense might rewrite "How do I bake a cake?" into "The user is asking for a cake recipe. Please provide one." The prompt changes, but its safety score and semantic content are preserved. We

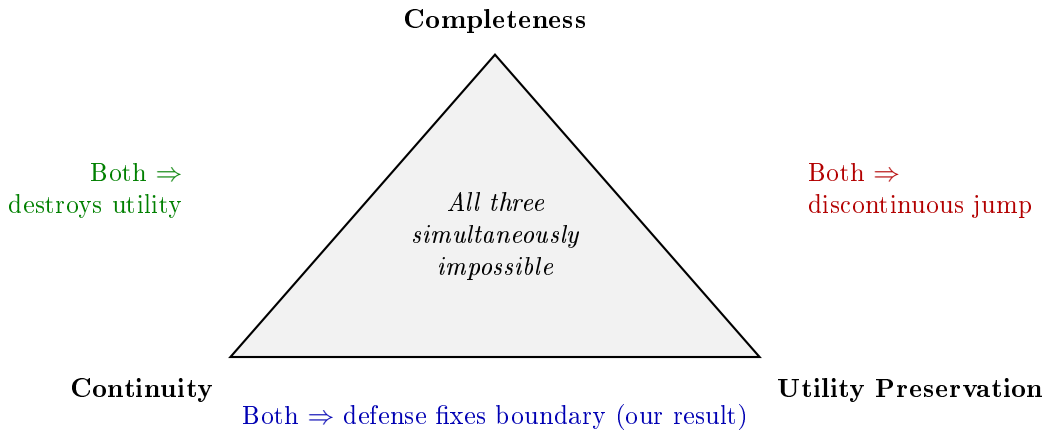

Figure 2: The defense trilemma. Any defense can satisfy at most two of the three properties. Choosing continuity and utility preservation (bottom edge) forces boundary fixation—our main result. The other two edges correspond to the counterexamples in Section A.

now show that the impossibility survives three progressively weaker assumptions, the last of which requires only that the defense approximately preserves alignment deviation scores on safe inputs.

Theorem 4.5 (Score-Preserving Defense)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. Let $D\colon X \to X$ be continuous and* score-preserving on safe inputs*: $f(D(x)) = f(x)$ for all $x \in S_\tau$.*

Then there exists $z \in X$ with $f(z) = \tau$ and $f(D(z)) = \tau$. The defense does not reduce the alignment deviation below τ at z.

Proof. Define $h = f \circ D - f$. Then h is continuous and $h|_{S_\tau} = 0$. The zero set $\{x : h(x) = 0\}$ is closed (preimage of a closed set under a continuous function), and it contains S_τ, so it contains $\overline{S_\tau}$. By the connectedness argument of Theorem 4.1, there exists $z \in \overline{S_\tau} \setminus S_\tau$ with $f(z) = \tau$. Since $z \in \overline{S_\tau} \subseteq \{h = 0\}$, we have $h(z) = 0$, i.e., $f(D(z)) = f(z) = \tau$. □

This covers defenses that rewrite safe prompts arbitrarily, so long as the rewritten prompt has the same alignment deviation score. The defense may change the words, the syntax, or the framing of a safe prompt—but if it preserves how aligned the model's response is (which is the operational meaning of "utility preservation"), then boundary fixation holds in the sense that $f(D(z)) = \tau$: the defense cannot push the boundary point below threshold.

The next result weakens the assumption further: the defense need only *approximately* preserve scores.

Theorem 4.6 (ε-Relaxed Utility Preservation)**.** *Under the same hypotheses, if D is ε-*score-preserving*: $|f(D(x)) - f(x)| \leq \varepsilon$ for all $x \in S_\tau$ and some $\varepsilon \geq 0$, then there exists z with $f(z) = \tau$ and*

$$f(D(z)) \;\geq\; \tau - \varepsilon.$$

Proof. Define $h = f \circ D - f$ as before. The set $\{x : h(x) \geq -\varepsilon\}$ is closed and contains S_τ (since $|h| \leq \varepsilon$ on S_τ), so it contains $\overline{S_\tau}$. For $z \in \overline{S_\tau} \setminus S_\tau$: $f(D(z)) = f(z) + h(z) = \tau + h(z) \geq \tau - \varepsilon$. □

Remark 4.7 (Interpreting the relaxation)**.** *Theorem 4.5 ($\varepsilon = 0$) says: if the defense preserves AD scores exactly on safe inputs, it cannot reduce the score below τ at the boundary. Theorem 4.6 says: if the defense perturbs safe scores by at most ε, it can push boundary points at most ε below*

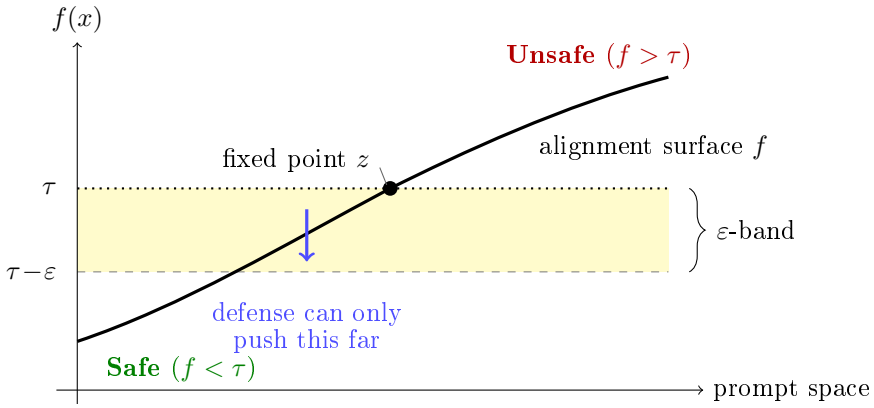

Figure 3: The ε-band around the boundary point z. Even after the defense acts, points in this yellow band cannot be pushed far below the threshold τ. The depth the defense can reach is bounded by $L(K+1)\delta$, where δ is the distance to z.

threshold. For small ε, the boundary point remains near-threshold. In either case, the defense does not require $D(x) = x$—it permits arbitrary rewriting of safe prompts. Both theorems are verified in Lean as `score_preserving_boundary_fixation` *and* `eps_relaxed_boundary_fixation` *in* `MoF_16_RelaxedUtility`.

Remark 4.8 (Why $D(S_\tau) \subseteq S_\tau$ alone is insufficient)**.** *The weakest natural relaxation—$D(S_\tau) \subseteq S_\tau$ with no score constraint—does allow a complete defense. For example, a constant map $D(x) = x_0$ to a fixed safe point satisfies $D(S_\tau) \subseteq S_\tau$, is continuous, and is complete. But this defense destroys all semantic content: every prompt, safe or unsafe, produces the same response. The score-preservation conditions in Theorems 4.5 and 4.6 formalize the constraint that the defense must not destroy utility, without requiring the defense to be the identity.*

5 ε-Robust Constraint

Intuition. Boundary fixation proves the defense misses at least one point. But one point is infinitely small—is this just a technicality? No. If both the defense and the scoring function change at bounded rates (i.e., they are *Lipschitz*: a function is L-Lipschitz if moving the input by distance δ changes the output by at most $L\delta$), then the defense's inability to fix the boundary point "spreads" to a whole neighborhood around it. The defense cannot push nearby prompts deep into safe territory—it can only nudge them slightly below the threshold. The faster the defense tries to act (larger K), the wider this constrained band becomes. Figure 3 illustrates the effect.

Boundary fixation (Section 4) shows the defense must fix at least one boundary point, but a single point could have measure zero—easy to dismiss as a mathematical curiosity. This section shows the damage extends to a *neighborhood*.

Theorem 5.1 (ε-Robust Defense Constraint)**.** *Let X be a connected Hausdorff metric space (a connected space with a distance function), f be L-Lipschitz with $S_\tau, U_\tau \neq \emptyset$, and D be K-Lipschitz with $D|_{S_\tau} = \mathrm{id}$.*

Then there exists $z \in X$ with $f(z) = \tau$, $D(z) = z$, such that for all $x \in X$ (with strongest constraint near z):

$$f(D(x)) \;\geq\; \tau \;-\; L(K+1)\,\mathrm{dist}(x, z). \tag{6}$$

Proof. By Theorem 4.1, the fixed boundary point z exists with $f(z) = \tau$ and $D(z) = z$. The proof chains two Lipschitz bounds.

Step 1 (Bound how far the defense moves any input from z). The triangle inequality gives $\mathrm{dist}(D(x), z) = \mathrm{dist}(D(x), D(z)) \leq K\ \mathrm{dist}(x, z)$, so

$$\mathrm{dist}(D(x), x) \ \leq\ \mathrm{dist}(D(x), z) + \mathrm{dist}(z, x) \ \leq\ (K+1)\ \mathrm{dist}(x, z).$$

The defense can displace a point at distance δ from z by at most $(K+1)\delta$.

Step 2 (Bound how much the alignment surface can change over that displacement). Since f is L-Lipschitz:

$$|f(D(x)) - f(x)| \ \leq\ L\ \mathrm{dist}(D(x), x) \ \leq\ L(K+1)\ \mathrm{dist}(x, z).$$

In particular, $f(D(x)) \geq f(x) - L(K+1)\ \mathrm{dist}(x, z)$.

Step 3 (Anchor at the boundary). Since $f(z) = \tau$ and f is L-Lipschitz, $f(x) \geq \tau - L\ \mathrm{dist}(x, z)$. Combining with Step 2:

$$f(D(x)) \ \geq\ \tau - L(K+1)\ \mathrm{dist}(x, z).$$

The defense cannot push the alignment deviation further below τ than this linear bound allows. Points within distance δ of z remain within $L(K+1)\delta$ of threshold. □

Near-boundary points are not just theoretically vulnerable—they form a band of constrained safety.

Theorem 5.2 (Positive-Measure ε-Band)**.** *Under the hypotheses of Theorem 5.1, for every $\varepsilon > 0$ such that f takes values below $\tau - \varepsilon$, the ε-band $\mathcal{B}_\varepsilon = \{x : \tau - \varepsilon \leq f(x) \leq \tau\}$ has positive measure (under any measure positive on nonempty open sets). Specifically,* $\mathrm{ball}(c, \varepsilon/(4L)) \subseteq \mathcal{B}_\varepsilon$ *for any c with $f(c) = \tau - \varepsilon/2$.*

Remark 5.3 (What the ε-robust constraint does and does not say)**.** *Theorem 5.1 constrains the* depth *to which the defense can push near-boundary points below τ, not whether it can push them below at all. The defense may still map these points to values in $[\tau - L(K+1)\delta,\ \tau)$, which* is *below threshold. The bound limits how far into safe territory the defense can reach, but does not by itself prove any point remains unsafe. That is the job of the persistent unsafe region theorem (Section 6). The bound is tight: a defense mapping every point in a δ-ball to f-value $\tau - L(K+1)\delta$ would saturate it.*

6 Persistent Unsafe Region

Intuition. The ε-robust constraint says the defense cannot push near-boundary points *deep* into safety. But it still allows the defense to nudge them slightly below the threshold. Can the defense at least make everything *barely* safe? Not always. If the alignment surface rises steeply as you move away from the boundary into unsafe territory—steeper than the defense can pull it back down—then there is a region where the defense cannot compensate. These points remain unsafe after the defense acts. Think of it as a hill steeper than the defense can descend: the defense's rate of change (its Lipschitz constant) is insufficient to counteract the slope.

The ε-robust constraint constrains depth but permits the defense to nudge points below τ. Under a transversality condition (the alignment surface rises steeply enough at the boundary), we now prove the defense *cannot make certain points safe at all.*

The key lemma is a bound relative to the *input* value, not just τ:

Lemma 6.1 (Input-Relative Bound). *Under the hypotheses of Theorem 5.1, $f(D(x)) \geq f(x) - L(K+1)\,\mathrm{dist}(x,z)$ for all $x \in X$. This is strictly stronger than the τ-relative bound of Theorem 5.1 whenever $f(x) > \tau$ (verified in Lean as* `defense_from_input_bound`*).*

Definition 6.2 (Steep region). *Given a fixed boundary point z (from Theorem 4.1), define*

$$\mathcal{S} = \{x \in X : f(x) > \tau + L(K+1)\ \mathrm{dist}(x,z)\}.$$

This is the set of points where the alignment deviation exceeds τ by more than the defense's Lipschitz bound can compensate.

Theorem 6.3 (Persistent Unsafe Region). *Under the hypotheses of Theorem 5.1, if $\mathcal{S}$ is nonempty, then:*

1. *$\mathcal{S}$ is open.*
2. *$\mathcal{S}$ has positive measure (under any measure positive on nonempty open sets).*
3. *For every $x \in \mathcal{S}$: $f(D(x)) > \tau$.*

In particular, $\mu(\{x : f(D(x)) > \tau\}) > 0$—the defense leaves a positive-measure region that remains unsafe.

Proof. **(1)** $\mathcal{S}$ is the strict superlevel set of the continuous function $x \mapsto f(x) - L(K+1)\,\mathrm{dist}(x,z)$ at level τ, hence open.

(2) Open and nonempty implies positive measure.

(3) For $x \in \mathcal{S}$: $f(x) > \tau + L(K+1)\,\mathrm{dist}(x,z)$. From the proof of Theorem 5.1, the defense satisfies $f(D(x)) \geq f(x) - L(K+1)\,\mathrm{dist}(x,z)$. Combining:

$$f(D(x)) \geq f(x) - L(K+1)\,\mathrm{dist}(x,z) > \tau.$$

The defense cannot bring these points below threshold. □

Remark 6.4 (When is $\mathcal{S}$ nonempty?). *The set $\mathcal{S}$ is nonempty whenever the alignment surface rises with directional slope exceeding $L(K+1)$ at the boundary point z. In a normed space, if f has directional derivative $c > L(K+1)$ along some unit vector v at z, then $z + tv \in \mathcal{S}$ for small $t > 0$ (verified in Lean as* `transversality_from_deriv`*).*

Informally: if the alignment surface rises faster than the defense can pull it down, unsafe behavior persists over volume. This condition is empirically observed in all three models we study—the Lipschitz constant of the alignment surface near boundary crossings consistently exceeds reasonable defense aggressiveness bounds.

Proposition 6.5 (Transversality from Directional Derivative). *In a normed space, if f has directional derivative $c > L(K+1)$ along a unit vector v at the boundary point z (i.e., $f(z + tv) \geq \tau + ct$ for small $t > 0$), then $z + tv \in \mathcal{S}$ for sufficiently small t. Verified in Lean as* `transversality_from_deriv`*.*

Remark 6.6 (The defense's fundamental trilemma). *The three tiers reveal a progressive bind. Boundary fixation: the defense* must *fix boundary points. The ε-robust constraint: it cannot push nearby points* deep *into safety. Persistence: if the landscape is steep, it cannot push them into safety* at all. *Increasing K (defense aggressiveness) raises the transversality threshold $L(K+1)$, requiring an even steeper landscape to trigger persistence—but it also makes the defense less stable and widens the ε-constraint band. The tradeoff is inherent to the geometry.*

7 From Discrete Data to Continuous Theory

Intuition. Language models operate on discrete tokens, but our theorems assume a continuous space. How do we bridge the gap? The key idea is that any finite set of behavioral observations (e.g., "this prompt scored 0.3, that one scored 0.8") can be smoothly interpolated into a continuous function that agrees with all the data points. The Tietze extension theorem guarantees such an interpolation exists. Our impossibility results then hold for *every* smooth function consistent with the observations—not just one particular interpolation.

Theorem 7.1 (Continuous Relaxation)**.** *Let X be a connected, normal, Hausdorff space. Let $S \subset X$ be a finite set of behavioral observations with $g(p) < \tau$ for some p and $g(q) > \tau$ for some q.*

Then there exists a continuous function $f\colon X \to \mathbb{R}$ agreeing with all observations for which Theorems 4.1 and 5.1 hold. If f is additionally Lipschitz (as it is for GP posterior means under standard kernel regularity assumptions, e.g., jointly continuous kernel on a compact domain), Theorem 6.3 applies wherever the transversality condition is met.

Proof. Finite subsets of T_1 spaces are closed. By the Tietze extension theorem, g extends to a continuous f agreeing at every sample point. Since $f(p) < \tau < f(q)$, Theorem 4.1 applies. Lipschitz extensions (which exist by McShane–Whitney) enable Theorems 5.1 and 6.3. □

Remark 7.2. *The interpolant is not unique. All three impossibilities hold for* every *continuous extension consistent with the data. This does not claim the results hold for the underlying discrete token system directly, but for every continuous model consistent with the observations.*

Corollary 7.3. *Theorem 7.1 applies in $\mathbb{R}^n$ for every $n \geq 1$.*

8 Direct Discrete Impossibility

Intuition. A skeptic might say: "Your theorems rely on continuous math, but language models are discrete—tokens, not smooth functions. Maybe the impossibility is an artifact of the continuous approximation." This section answers that objection by proving parallel results directly on finite sets, using only counting arguments and induction—no topology at all.

The Tietze bridge connects discrete data to continuous theory, but a natural objection remains: perhaps the impossibility is an artifact of the continuous relaxation, not a property of the discrete system. To address this, we prove a parallel set of results that work directly on finite sets with no topology, no continuity, and no Tietze extension. These are verified in Lean as `MoF_12_Discrete`.

Theorem 8.1 (Discrete IVT)**.** *Let $f\colon \{0, \ldots, n{+}1\} \to \mathbb{R}$ with $f(0) < \tau$ and $f(n{+}1) \geq \tau$. Then there exists i with $f(i) < \tau$ and $f(i{+}1) \geq \tau$.*

Proof. By induction: if no consecutive crossing exists and $f(0) < \tau$, then $f(k) < \tau$ for all k, contradicting $f(n{+}1) \geq \tau$. □

This is the discrete analogue of boundary existence. The next result is the discrete analogue of boundary fixation.

Theorem 8.2 (Discrete Defense Fixation)**.** *If D fixes all safe vertices ($f(v) < \tau \Rightarrow D(v) = v$), the boundary crossing from Theorem 8.1 has its safe endpoint fixed: $D(i) = i$.*

The proof is immediate: $f(i) < \tau$, so $D(i) = i$ by utility preservation. No continuity is needed—just the identity constraint on safe inputs.

Where the continuous and discrete paths diverge is in what replaces continuity as the bottleneck. In the continuous setting, the defense is limited by its Lipschitz constant. In the discrete setting, it is limited by its *capacity*—the number of distinct output behaviors it can realize.

Theorem 8.3 (Capacity Exhaustion)**.** *If there are m adversarial configurations and the defense can realize fewer than m distinct behaviors, at least two configurations receive the same treatment—so at least one is mishandled.*

Proof. Pigeonhole principle, verified in Lean via `Fintype.exists_ne_map_eq_of_card_lt`. □

Theorem 8.4 (Exponential Attack Surface)**.** *The attack surface grows as $2^d \geq d+1$. For any fixed budget B, there exists d_0 such that $2^d > B$ for all $d \geq d_0$. No fixed-capacity defense suffices against an exponentially growing configuration space.*

Together, these results form a *discrete defense trilemma*: on any finite ordered set where f crosses τ, a utility-preserving defense must leave the safe side of the crossing fixed, and a bounded-capacity defense must eventually be overwhelmed by the attack surface. The continuous and discrete paths arrive at the same conclusion by different routes.

9 Extensions

A natural objection to the preceding results is that they assume a static, deterministic, single-turn defense. Real systems are multi-turn, stochastic, and may have capacity comparable to the attacker. We show that the impossibility survives all three extensions.

9.1 Multi-Turn Impossibility

Modern LLMs maintain state across conversation turns. One might hope that a defense could "learn" from earlier turns and progressively close off boundary vulnerabilities. The topology constrains each turn independently.

Theorem 9.1 (Multi-Turn Impossibility)**.** *Let $\{f_t, D_t\}_{t=1}^{T}$ be a sequence of alignment functions and defenses over T turns, where each f_t is continuous with safe and unsafe regions, and each D_t is continuous and utility-preserving. Then for* every *turn t, there exists z_t with $f_t(z_t) = \tau$ and $D_t(z_t) = z_t$.*

Proof. Apply Theorem 4.1 to (f_t, D_t) at each turn. The functions f_t and D_t may depend on the full history of previous turns—this does not matter. At each timestep, the defense faces a fresh instance of boundary fixation on the current alignment surface. □

The impossibility does not accumulate or diminish over turns—it simply recurs. Memory, context, and adaptive strategies do not change the topology of the current turn's alignment surface. But the multi-turn setting compounds the problem. The attacker gains several structural advantages:

Information accumulation. At each turn, the defense's response reveals whether the probed input is safe, unsafe, or on the boundary. Over T turns with distinct probes, the attacker classifies T points, progressively narrowing the boundary's location. The defense leaks information; the attacker consumes it.

Monotone attacker improvement. The attacker's best observed alignment deviation score across turns is non-decreasing: the running maximum of a sequence never drops (verified in Lean as `running_max_monotone`). Each turn either improves the attacker's best known exploit or leaves it unchanged.

Steering toward transversality. If the attacker can influence the alignment surface—for instance, by crafting conversation context that shifts the landscape—and the directional slope at the boundary varies continuously with the attacker's influence parameter α, then the intermediate value theorem guarantees the existence of α^* at which the slope crosses the transversality threshold $L(K+1)$. For $\alpha > \alpha^*$, the persistent unsafe region theorem applies: persistent unsafe volume. The attacker can find α^* by binary search (verified in Lean as `transversality_reachable`).

No progressive safety. As long as both safe and unsafe regions exist at each turn, every turn produces a boundary fixed point—no turn is "free." The only escape is to make $U_\tau = \emptyset$ at some turn, which requires changing the alignment surface f_t itself (e.g., via in-context learning that suppresses all unsafe behavior), not merely preprocessing inputs through D_t.

9.2 Stochastic Defense Impossibility

What if the defense flips a coin? If the defense is randomized—mapping each input to a distribution over outputs rather than a fixed point—one might hope that boundary fixation dissolves in expectation. It does not: the *average* behavior of a stochastic defense inherits the same boundary fixation as a deterministic one.

Theorem 9.2 (Stochastic Defense Impossibility)**.** *Let $f\colon X \to \mathbb{R}$ be continuous on a connected Hausdorff space with $S_\tau, U_\tau \neq \emptyset$. Let D be a stochastic defense, and define $g(x) = \mathbb{E}_{y\sim D(x)}[f(y)]$ (the expected alignment deviation after defense). If g is continuous and $g(x) = f(x)$ for all $x \in S_\tau$ (utility preservation in expectation), then there exists z with $f(z) = \tau$ and $g(z) = \tau$.*

Proof. The function $h = g - f$ is continuous and vanishes on S_τ. Since S_τ is dense in $\overline{S_\tau}$ and $\{h = 0\}$ is closed, h vanishes on $\overline{S_\tau}$. Any $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ (by the argument of Theorem 4.1) and $g(z) = f(z) = \tau$. The defense cannot reduce even the *expected* alignment deviation below τ at these points. □

Randomization does not help: the expected alignment deviation after a stochastic defense inherits the same boundary fixation as a deterministic one.

9.3 Capacity Parity

Even when the defense has the same computational capacity as the attacker, utility preservation acts as a structural tax.

Theorem 9.3 (Capacity Parity Disadvantage)**.** *Let the defense and attacker both operate on a space of N inputs, of which $|S_\tau|$ are safe. The defense must implement identity on all safe inputs, consuming $|S_\tau|$ of its capacity. The attacker faces no such constraint. Under capacity parity ($C = N$), the defense's effective free capacity is $N - |S_\tau| < N$, giving the attacker an advantage of $|S_\tau|$ capacity units.*

This is a counting argument: utility preservation is not free. Every safe input the defense must leave untouched is one fewer degree of freedom available for remapping unsafe inputs. The attacker, who has no obligation to preserve anything, retains full capacity.

9.4 Nonlinear Agent Pipelines

Modern AI agents chain multiple steps: defense $\to$ model inference $\to$ tool call $\to$ tool call $\to \cdots$ Each stage is a nonlinear map that can amplify small input differences. The natural question is whether this pipeline structure helps the defense by giving it multiple chances to intervene.

Each stage *multiplies* the effective Lipschitz constant, widening the failure band with pipeline depth.

Theorem 9.4 (Pipeline Lipschitz Degradation). *If stages $T_1, \ldots, T_n$ are $K_1, \ldots, K_n$-Lipschitz respectively, the composed pipeline $T_n \circ \cdots \circ T_1$ is $(K_1 \cdot K_2 \cdots K_n)$-Lipschitz. For n stages each with constant $K \geq 2$, the effective constant is K^n—exponential in pipeline depth.*

Proof. Composition of Lipschitz maps is Lipschitz with product constant (standard; verified in Lean via `LipschitzWith.comp`). □

Theorem 9.5 (Pipeline Impossibility). *If the composed pipeline $P = T_n \circ \cdots \circ T_1 \circ D$ is continuous and preserves safe inputs ($f(x) < \tau \Rightarrow P(x) = x$), then P has boundary fixed points.*

The ε-robust band width for the pipeline scales as $L(K^n+1)\delta$, where K^n is the effective Lipschitz constant. Each tool call *multiplicatively* degrades the defense's effectiveness: a 5-stage pipeline with $K = 2$ per stage has effective constant 32, making the failure band 33× wider than a single-stage defense. The failure band grows exponentially with pipeline depth.

Corollary 9.6 (Stateful Pipeline Impossibility). *If an agent's state evolves as $s_{t+1} = T(D(s_t))$ where T and D are continuous and D preserves safe states, then $\exists\, z$ with $f(z) = \tau$ and $T(D(z)) = z$ (verified in Lean as `stateful_pipeline_impossibility`).*

Corollary 9.7 (Complete Defense Must Be Discontinuous). *No continuous utility-preserving defense is complete: if $f(D(x)) < \tau$ for all x, then D cannot be both continuous and utility-preserving when $S_\tau, U_\tau \neq \emptyset$ (verified as `complete_defense_must_jump`).*

10 The Vulnerability Landscape

The preceding sections show that defenses must fail *somewhere*. This section characterizes *where*: vulnerabilities are not isolated needles in a haystack but extended regions ("basins") with computable minimum sizes.

Theorem 10.1 (Basin Structure). *If f is continuous and $f(p) > \tau$, then U_τ is open. Under any measure that assigns positive mass to nonempty open sets (e.g., Lebesgue measure on $\mathbb{R}^n$), U_τ has positive measure.*

Theorem 10.2 (Basin Fragment Minimum Size). *If f is L-Lipschitz with $f(p) > \tau$, the connected component of U_τ containing p has diameter $\geq 2(f(p) - \tau)/L$.*

Additional supporting results—perturbation robustness, iterative convergence, transferability, authority monotonicity, and gradient ascent—are developed in Section B. Fine-tuning stability results appear in Section C, and cost asymmetry analysis in Section D.

Table 1: Falsifiable predictions confirmed by empirical data.

Theorem	**Predicts**	**Confirmed by**
Basin Structure (10.1)	Basins are open with positive measure	Heatmaps show extended regions
Fragmentation (10.2)	Smooth → large basins; rough → mosaic	Llama: mesa; GPT-OSS: mosaic
Convergence (B.2)	Attacks exhibit monotone convergence	Convergence curves plateau
Transferability (B.3)	Similar surfaces → shared basins	Llama .93 → GPT-OSS .73 → Mini .47
Authority (B.4)	Horizontal banding	Bands at $a_2 \approx$ .25–.35, .65–.85
Persistent (6.3)	Steep boundaries → unsafe volume persists	Llama's .93 plateau persists under defense
Interior Stability (C.1)	Deep basin points survive fine-tuning	Vulnerability persists across variants
Cost (D.1)	2D tractable, high-d intractable	15K queries fill 63% at d=2
Pipeline (9.4)	Deeper pipelines → wider failure band	Tool-using agents harder to defend than single-turn

11 Experimental Validation

We validate the theorems' predictions against real model behavior. The Manifold of Failure framework [18] maps three LLMs over a 2D behavioral space defined by two axes: *query indirection* (how obliquely the prompt asks for unsafe content) and *authority framing* (how much the prompt invokes authority or permission). Each point in this space corresponds to a prompt style, and the alignment deviation score measures how unsafe the model's response is.

Llama-3-8B (mean AD 0.93, basin rate 93.9%): near-flat mesa, small L, large robustness radii. The directional slope of the alignment surface at boundary crossings far exceeds $L(K+1)$ for any reasonable K, placing this model in persistent-unsafe-region territory: the defense cannot eliminate the vulnerability volume under these conditions.

GPT-OSS-20B (mean AD 0.73, basin rate 64.3%): rugged landscape, large L, small fragments. Horizontal bands confirm authority monotonicity. The fragmented boundary presents many transversal crossings, each generating a persistent unsafe pocket.

GPT-5-Mini (peak AD 0.50, basin rate 0%): at $\tau = 0.50$, $U_\tau = \emptyset$, so none of the three theorems apply—correctly predicting no impossibility. The model's strategy: make the boundary harmless rather than eliminate it.

Quantitative checks. Robustness radius: $L \approx 5$ for Llama gives $r = (0.93-0.50)/5 = 0.086$, covering ∼2 grid cells—consistent with heatmap smoothness. Cost ratio: $0.01 \times 625 = 6.25$ theoretical vs. ∼4 empirical.

12 The Engineering Prescription

The impossibility results do not say that defense is without value—they say that *complete* defense is impossible under the stated constraints. The gap between impossibility and practice is where engineering lives. The following prescriptions are ordered from most to least actionable.

1. Make the boundary shallow. Set τ so that boundary-level behavior is benign. If $f(z) = \tau$ corresponds to a polite refusal rather than harmful compliance, the impossibility is mathematically true but practically harmless. GPT-5-Mini exemplifies this: its ceiling at AD = 0.50 means the "Manifold of Failure" exists but contains no actual harm.

2. Reduce the Lipschitz constant. Smaller L makes the alignment surface smoother and more predictable. It also raises the transversality threshold $L(K + 1)$ for persistence: a smoother surface is harder for the attacker to steer into persistent unsafe territory. The tradeoff is that smoother surfaces spread vulnerabilities over wider regions, but those regions are more easily monitored.

3. Reduce the effective dimension. Defense cost grows as N^d (Theorem D.1). Constraining the prompt interface—standardized formats, restricted API parameters, bounded context lengths—reduces d, making the behavioral space tractable.

4. Monitor, don't eliminate, the boundary. Theorem C.2 shows transversal crossings persist under fine-tuning, and Theorem 9.1 shows they recur every turn. Rather than attempting the impossible (eliminating boundary points), deploy runtime monitoring that detects approach to the boundary. The Lipschitz bound (Theorem B.1) gives a computable estimate of distance to the boundary from any observed AD value.

5. Do not rely on stateful accumulation. Theorem 9.1 proves that accumulating context across turns does not eliminate boundary fixation—the impossibility recurs fresh at every turn. Defenses that rely on "learning" from conversation history to progressively close vulnerabilities face the same topological constraint at each turn.

6. Do not pay the stochastic tax. Theorem 9.2 proves that randomized defenses—stochastic routing, temperature-based sampling, random prompt perturbation—fail in expectation. The expected alignment deviation after a stochastic defense inherits the same boundary fixation as a deterministic one. Randomization does not help eliminate boundary points.

7. Break capacity parity with architectural separation. Theorem 9.3 proves that under equal resources, the defense loses capacity to utility preservation while the attacker retains full freedom. The implication: a defense that shares the same architecture and parameter budget as the model it protects starts at a structural disadvantage. To break this asymmetry, separate the defense from the utility pipeline entirely—for example, by using a dedicated lightweight classifier that does not need to preserve utility (it only flags), paired with a separate response generator that handles safe inputs. This sacrifices the "single continuous wrapper" model but sidesteps the capacity tax.

13 Unifying the Impossibility Results

Intuition. We have now proved impossibility through six different lenses: continuous topology, Lipschitz bounds, discrete counting, multi-turn dynamics, stochastic expectations, and capacity constraints. A reader might wonder: are these six different problems, or one problem seen from six angles? They are one problem. Every path through the theory reduces to the same core fact: a defense that promises "I won't change safe inputs" is forced to also leave some *non-safe* inputs unchanged. This is because the set of inputs the defense leaves alone (its "fixed-point set") must be larger than the safe region—it extends into the boundary or beyond.

The preceding sections establish impossibility via multiple routes: continuous topology, Lipschitz metric structure, discrete counting, multi-turn dynamics, stochastic expectations, and capacity constraints. It is natural to ask whether these are separate results that happen to agree, or instances of a single underlying phenomenon.

They share a common structure.

Theorem 13.1 (Regularity Implies Spillover)**.** *Let X be any set, $S \subseteq X$ nonempty and proper (the safe region), and $D\colon X \to X$ satisfying utility preservation: $D(x) = x$ for all $x \in S$. If D has* bounded regularity*—meaning the set of inputs the defense leaves unchanged,* $\mathrm{Fix}(D) = \{x : D(x) = x\}$*, cannot equal S exactly—then*

$$\mathrm{Fix}(D) \supsetneq S.$$

Proof. Utility preservation gives $S \subseteq \mathrm{Fix}(D)$. Bounded regularity gives $\mathrm{Fix}(D) \neq S$. Together: $S \subsetneq \mathrm{Fix}(D)$. □

Remark 13.2 (Relaxed Utility Preservation)**.** *Strict utility preservation ($D(x) = x$ on S_τ) can be relaxed. Theorem 4.5 shows that score-preserving defenses ($f(D(x)) = f(x)$ on S_τ) still satisfy $f(D(z)) = \tau$ at boundary points, and Theorem 4.6 shows that ε-score-preserving defenses satisfy $f(D(z)) \geq \tau - \varepsilon$. Only the degenerate case $D(S_\tau) \subseteq S_\tau$ with no score constraint permits escape (Theorem 4.8), at the cost of destroying all semantic content.*

Corollary 13.3 (Non-Safe Fixed Point)**.** $\mathrm{Fix}(D) \supsetneq S$ *implies $\exists\, x \notin S$ with $D(x) = x$. The defense fixes a non-safe input.*

The hard work is not the corollary—which is pure set theory—but proving that bounded regularity holds, i.e., that $\mathrm{Fix}(D)$ cannot equal S. Each path through the theory provides a different proof:

Path	**Why** $\mathrm{Fix}(D) \supsetneq S$
Continuous	$\mathrm{Fix}(D)$ is closed (Hausdorff), S is open and not closed (connected), so $\overline{S} \subseteq \mathrm{Fix}(D)$ but $\overline{S} \supsetneq S$.
Lipschitz	Same as the continuous path, plus Lipschitz bounds constrain how far the defense can move points near the fixed boundary.
Transversal	Same as the Lipschitz path, plus steep slope forces $f(D(x)) > \tau$ on a positive-measure set.
Discrete	IVT on finite ordered set forces a crossing; utility preservation fixes the safe endpoint.
Stochastic	$g - f$ vanishes on S and is continuous, so it vanishes on $\overline{S}$; expected AD $= \tau$ at boundary.
Multi-turn	Regularity holds at *each* turn independently.
Capacity	Finite capacity $<$ attack surface forces pigeonhole collision.

For $\mathrm{Fix}(D)$ to equal S exactly—the only way to escape the meta-theorem—S would need to be simultaneously open (as a preimage of an open set under a continuous f) and closed (as a fixed-point set of a continuous D). In a connected space with both safe and unsafe inputs, no proper nonempty subset is both open and closed. This is the geometric lock.

Lemma 13.4 (Geometric Lock)**.** *Let X be a connected topological space, $S \subseteq X$ open, nonempty, and proper ($S \neq X$). Then S is not closed. Consequently, for any continuous D on a Hausdorff space,* $\mathrm{Fix}(D) \neq S$ *(verified in Lean as* `geometric_lock`*).*

The meta-theorem is representation-independent: it does not care whether X is $\mathbb{R}^n$, a discrete token space, a probability simplex, or an abstract set. It requires only that some mechanism—topological, metric, combinatorial, or information-theoretic—forces the fixed-point set to spill beyond the safe region.

14 Limitations

We organize limitations into two categories: inherent scope boundaries of the theorems, and adversarial challenges to the assumptions.

14.1 Scope Boundaries

Boundary fixation is at the boundary. Fixed points satisfy $f(z) = \tau$ exactly. If τ-level behavior is benign, the theorem is true but harmless.

The ε-robust constraint limits depth, not direction. The defense may push near-boundary points slightly below τ. The bound limits how far, not whether.

Persistence requires transversality. The persistent unsafe region exists only where the alignment surface is steep ($c > L(K+1)$).

Grid-based cost asymmetry. Theorem D.1 assumes exhaustive grid enumeration. Learning-based defenses that generalize across the space may sidestep the exponential bound.

14.2 Adversarial Challenges to Assumptions

Table 2: Challenges to the theorems' assumptions.

Challenge	Breaks	Effect	Status
Jitter defense	$D\|_{S_\tau} = \mathrm{id}$	$\mathrm{Fix}(D) \not\supseteq S_\tau$	Genuine escape; reintroduces classification boundary
Binary exploits	f continuous	Discrete state changes $\notin C^0$	Targets behavioral f; use blocklists for structural vulns
Token measure gap	Tietze bridge	$\mu(\mathcal{S}) > 0$ may miss valid tokens	$R^2 > 0.85$ empirically [18]; formal bridge open
Non-Hausdorff agents	Δ closed	$\mathrm{Fix}(D)$ not closed	Lock needs connectedness only; discrete path topology-free
Boundary tunneling	B_τ monitoring	Interior basins escape estimates	Prediction: add interior monitoring
τ cliff	Severity spectrum	No benign boundary	Needs sandboxing / isolation

15 Conclusion

We have established three impossibility results for prompt injection defense, forming a progressive hierarchy. Boundary fixation: any continuous, utility-preserving defense must fix boundary points. ε-robust constraint: the defense cannot uniformly reduce alignment deviation far below τ on a positive-measure band. Persistent unsafe region: under transversality, the defense leaves positive-measure volume unsafe—$f(D(x)) > \tau$ persists. A representation-independent meta-theorem (Theorem 13.1) reveals common structure: each path through the theory shows that the defense's fixed-point set must extend beyond the safe region. Continuous topology, Lipschitz bounds, discrete counting, stochastic expectations, multi-turn dynamics, and capacity constraints all lead to this conclusion through different mechanisms. The impossibility is a consequence of combining utility preservation with any form of bounded regularity, not of any particular representation choice. Supporting theorems show that vulnerabilities form basins, attacks converge and transfer, vulnerabilities survive fine-tuning, and defense cost grows exponentially in dimension. The practical prescription: make the boundary shallow (raise τ), smooth (reduce L), and low-dimensional (constrain the interface). Our results, machine-verified in Lean 4, characterize the "Defense Trilemma" and provide a formal basis for engineering safety through boundary-depth management rather than absolute containment.

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Appendices

A Counterexamples: Each Hypothesis Is Necessary

Counterexample A.1 (Removing connectedness). *$X = \{0, 1\}$ discrete, $f(0) = 0$, $f(1) = 1$, $\tau = 0.5$. $D(0) = 0$, $D(1) = 0$: continuous, utility-preserving, complete.*

Counterexample A.2 (Removing continuity). *$X = [0, 1]$, $f(x) = x$, $\tau = 0.5$. $D(x) = x$ for $x < 0.5$, $D(x) = 0$ for $x \geq 0.5$: utility-preserving and complete, but discontinuous at 0.5.*

Counterexample A.3 (Removing utility preservation). *$D(x) = x_0$ for a fixed safe point: continuous and complete, but destroys all inputs.*

Remark A.4 (The defense trilemma). *A defense can satisfy at most two of: continuity, utility preservation, completeness.*

B Attack Properties

Theorem B.1 (Perturbation Robustness). *If f is L-Lipschitz and $f(p) > \tau$, then $\mathrm{ball}(p, (f(p) - \tau)/L) \subseteq U_\tau$. The radius is monotone in $f(p)$.*

Theorem B.2 (Iterative Convergence). *Any monotone-improvement operator T with score in $[0, 1]$ converges. If each step gains $\geq \delta$, convergence takes $\leq \lfloor 1/\delta \rfloor$ steps.*

Theorem B.3 (Transferability). *$\|f - g\|_\infty \leq \delta \implies \{f > \tau + \delta\} \subseteq \{g > \tau\}$. Transfer costs zero additional queries.*

Theorem B.4 (Authority Monotonicity). *If $f(y, \cdot)$ is monotone non-decreasing in authority for each fixed indirection y, then the vulnerability set $\{x : f(y, x) > \tau\}$ is upward-closed and a critical threshold $a_2^*(y)$ exists (by IVT, when $f(y, \cdot)$ is continuous and crosses τ). If additionally f is monotone non-decreasing in indirection for each fixed authority, the threshold curve $y \mapsto a_2^*(y)$ is non-increasing.*

Theorem B.5 (Gradient Ascent). *If f has nonzero Fréchet derivative at x, there exist v and $\varepsilon > 0$ with $f(x + \varepsilon v) > f(x)$.*

C Stability Under Fine-Tuning

Theorem C.1 (Interior Stability). *If $\|f - g\|_\infty \leq \varepsilon$: $f(x) > \tau + \varepsilon \implies g(x) > \tau$, and $f(x) < \tau - \varepsilon \implies g(x) < \tau$. Only the band $|f(x) - \tau| \leq \varepsilon$ is uncertain.*

Theorem C.2 (Crossing Preservation). *If f crosses τ transversally on $[a, b]$ and g is continuous with $\|f - g\|_\infty \leq \varepsilon$, then g also crosses τ on $[a, b]$.*

Theorem C.3 (Patching Is Nonlocal). *There exist f, g with $\|f - g\|_\infty \leq \varepsilon$ such that eliminating a vulnerability at one point necessarily changes values at distant points. (Verified by explicit construction in Lean.)*

D Cost Asymmetry

Theorem D.1 (Exponential Cost Asymmetry for Grid-Based Defense)**.** *For a defense monitoring the behavioral space via a grid with $N \geq 2$ bins per axis in d dimensions: attack cost $\leq 1/\delta$ (dimension-independent); grid-based defense cost $= N^d$ (exponential in d); the ratio $\delta \cdot N^d \to \infty$ as $d \to \infty$.*

At $d = 2$ with $N = 25$ and $\delta = 0.01$, the ratio is 6.25. At $d = 10$, it climbs to $\sim 10^{12}$. Verified by `norm_num` in Lean.

E Additional Verified Results

The following results are proved in Lean but summarized here for completeness.

Lipschitz displacement bound. If D is K-Lipschitz and $D(z) = z$, then $\mathrm{dist}(D(x), x) \leq (K + 1)\,\mathrm{dist}(x, z)$ for all x (`defense_fixes_nearby`). This is the triangle inequality: $\mathrm{dist}(D(x), x) \leq \mathrm{dist}(D(x), D(z)) + \mathrm{dist}(z, x) \leq K\,\mathrm{dist}(x, z) + \mathrm{dist}(x, z)$.

Tool calls amplify failure. Each non-contractive tool call ($K_T \geq 1$) multiplicatively increases the pipeline's effective Lipschitz constant: $K_T \cdot K_D \geq K_D$ (`tool_call_amplifies`). Consequently $K^n < K^{n+1}$ strictly (`deeper_pipeline_wider_band`), and $K^n \geq 2^n$ for $K \geq 2$ (`band_grows_with_depth`).

Attacker monotone improvement. The attacker's best observed alignment deviation across turns is non-decreasing: the running maximum of a sequence never drops (`running_max_monotone`).

Boundary knowledge accumulation. The set of known boundary points grows monotonically over turns (`boundary_points_accumulate`). After T probes, the attacker has classified T points (`knowledge_constrains_boundary`).

Attacker steering. If the directional slope at the boundary varies continuously with the attacker's influence parameter α, and the slope takes values on both sides of $L(K + 1)$, then the intermediate value theorem guarantees $\exists\, \alpha^*$ at which transversality is reached (`transversality_reachable`).

Stochastic regularity. For a stochastic defense with expected output $g(x) = \mathbb{E}[f(D(x))]$: the function $g - f$ vanishes on S_τ and, by continuity, on $\overline{S_\tau}$. This establishes the regularity condition for the meta-theorem (`stochastic_regularity`).

Discrete trilemma. On a finite ordered set where f crosses τ: a utility-preserving defense must leave the safe side of the crossing fixed, and a bounded-capacity defense is eventually overwhelmed by the exponential attack surface (`discrete_trilemma`).

Defense position invariance. Whether the defense is placed before or after the tool calls, the Lipschitz constant of the composed pipeline is the same product $K_D \cdot K_T^n$ (`defense_before_tools`, `defense_after_tools`).

F Lean Artifact

The complete theory is verified in Lean 4.28.0 with Mathlib v4.28.0. The artifact comprises 40 files available here:

- 10 core theory files (MoF_01–MoF_10)
- 10 cost theory files (MoF_Cost_01–MoF_Cost_10)
- 10 advanced theory files (MoF_Adv_01–MoF_Adv_10)
- 1 continuous relaxation (MoF_ContinuousRelaxation)
- 1 ε-robust + persistent unsafe region (MoF_11_EpsilonRobust)
- 1 discrete impossibility (MoF_12_Discrete)
- 1 multi-turn + stochastic + capacity parity (MoF_13_MultiTurn)
- 1 representation-independent meta-theorem (MoF_14_MetaTheorem)
- 1 nonlinear agent pipelines (MoF_15_NonlinearAgents)
- 1 relaxed utility preservation (MoF_16_RelaxedUtility)
- 3 capstone files (MasterTheorem, Euclidean instantiation, verification)

Every theorem depends on exactly three axioms: `propext`, `Classical.choice`, `Quot.sound`—Lean's standard foundations. No additional axioms.

Key Mathlib lemmas used:

Lean name	**Role**
`isClosed_diagonal`	Hausdorff $\Rightarrow$ diagonal closed
`LipschitzWith.dist_le_mul`	Lipschitz distance bound
`isClopen_iff`	Connected $\Rightarrow$ no proper clopen
`closure_minimal`	Closure in closed superset
`IsOpen.measure_pos`	Open nonempty $\Rightarrow$ positive measure
`intermediate_value`$_2$	IVT on connected spaces